

When forecasting the all-transaction house price index for California, I have used six different approaches in doing so. The first of these methods is the Naïve One forecast and it is straight forward in how it is created. It uses the last observed value as the new forecast. There are five different measures of accuracy that are used for this analysis but all aim to see how close the forecast is to the observed value. The smaller the values of these measurements, the better the forecast. When applied to Naïve One, it's more of a reference point for further forecasts.

Naïve Two is similar but adds what is called a change factor into the calculation. It would be a parameter that is between 0 and 1. With that, we take the difference of the last two observations and add it to the last observation. This results in a forecast value for tomorrow and is the same for any further days of interest. The accuracy of this forecast when compared to Naïve One improves greatly. The values are nearly half of what is presented in Naïve One which makes the forecast more reliable.

The next method is the Weighted Moving Average, and it multiplies a specified number of values from the past with a set value that adds up to one. In the forecast, I wanted to have four values ahead, so I used the last four values of the past. I then multiplied it with four weights that all add up to one. Those values were chosen by a program in excel and only made the fourth weight set to one with all others zero. I thought it was strange, but it insists it is the right set of values for this calculation. When measuring the accuracy of this forecast, it has performed the worst so far. The values are a lot higher, which tells me that the difference between the forecasted value and the observed is drastic.

The Simple Exponential Smoothing method is used next, and it adds a new variable called the local mean level. It takes the average of the data points at the observed time and adjusts with further observations. A parameter that is used in the local mean level is calculated by Excel to give me the lowest value for the best result. However, it has given me a value of 1, which is odd and tells me that it can't go lower given the data set. The result from calculating the local mean level for the observed value is used to forecast the next day. When seeing how it performed, it has not done well as the values are higher than Naïve Two but does slightly better than the Weighted Moving Average.

Now, I'll use Holt's Exponential Smoothing to forecast the data and it adds another equation into the mix. The local level mean mentioned above is added with this new calculation that takes account the trends that appear in the data set. It also brings a new parameter to smooth out the forecast, and it is called "Beta". When measuring its accuracy, is nearly neck in neck with the Naïve Two forecast. Which tells me that the forecasts are close to the actual values themselves and could be a reliable way to predict the next few days.

The Holt-Winters model builds upon the last forecast and considers the seasonal occurrences that happen in the data set. The data set has two ways of calculating itself in that it depends on how the seasons behave over time. A consistent seasonal trend that has the same range of values occurring over time is turned into an additive model. If the seasonal trend is getting larger over time, then we use the multiplicative model. In the data set, I went with the multiplicative model as the seasons have been getting larger over time. When measuring its accuracy, it does slightly worse than the Holt's Exponential Smoothing

model. Which tells me that the seasonal component that is added to the calculation struggles in closing the gap.

Now, seeing how they all performed, the best method that was used is Naïve Two. Holt's Exponential method comes close but has a higher mean square error. That is a measurement of the average differences between the observed and forecasted value. The Naïve Two also has a lower Theil's U measurement. That one is a ratio of the current method and the Naïve One method. If the number is lower than 1, it is better than the Naïve One. In which Naïve Two is lower than all the other models in comparison.

The time plot in the next page shows the whole data set with the forecasted Naïve Two values and its 95% intervals at the end. The forecasted value from the best model expects a stagnant price for housing in the future quarters ahead. This is also consistent with Holt's Exponential model and expects seasonal variation to be minimal. Overall, I am confident that this forecast can be used for planning out the next four quarters.

